

SEASON 1 · EPISODE 1 · ENGLISH LOCALIZATION

Rise of the Raven

Hungarian and Serbian to American English: dubbing with voice cloning and frame-accurate lip sync, built on our custom workflow. Process and delivery report.

58	8,874	20+	52	100%
CLIPS DELIVERED	FRAMES ANALYZED	SPEAKING ROLES VOICED	LIP-SYNC PASSES	FRAME-EXACT & VERIFIED

01 — PROJECT OVERVIEW

Every line in English, in the actor's own voice, on the actor's own lips.

The objective

Convert the first episode of *Rise of the Raven* into American English for an international audience. Every character should keep the tone and character of the original actor's voice, and every on-camera line should be lip-synced so the English reads as if it had been performed on set.

The material

58 picture-locked clips of the episode (1920×1080, 23.976 fps) with their dialogue tracks. Most of the episode is in Hungarian, but the scenes with the Serbian lord Branković and his family are performed in Serbian, so the work covered two source languages.

What we accomplished

- All 58 clips delivered in American English, in the original order and with the original file names (plus an **_EN** suffix).
- Each speaking role voiced with a clone of its own actor's performance. Six principal characters received dedicated, purpose-built voice models.
- Lip sync on every shot where the speaker's face is visible, pinned to that speaker only. Shots with two people talking were handled in two passes.
- Every line checked against the original and corrected where needed: names, period vocabulary and meaning.
- Picture untouched wherever no lip sync was needed. Every file is frame-exact to its source.

58
Clips localized, 6 min 10 s of footage

2
Source languages: Hungarian and Serbian

20+
Speaking roles identified and voice-cloned

6
Dedicated character voice models

38
Clips lip-synced, 4,500+ frames re-rendered

52
Face-pinned lip-sync passes, incl. 3 two-speaker shots

36
Targeted script and audio corrections

58/58
Clips passed frame, sync and language checks

OUR APPROACH

Off-the-shelf dubbing gets a scene roughly right. Our custom workflow adds the precision a drama needs: it identifies who is speaking in every frame, keeps each character's voice consistent across the episode, verifies the English line by line against the original, and only alters the picture where the speaker's mouth is actually on screen.

02 — OUR WORKFLOW

Eight stages, each with its own checks

1

Scene, shot and face analysis

Every frame is scanned: shot boundaries detected, every face found, tracked across the shot and mapped with a dense facial-landmark mesh.

Shot detection · Face detection & tracking · 478-point landmark mesh

2

Speaker identification

Each line is matched to the person saying it, using mouth movement measured frame by frame plus voice analysis of every sentence. On-screen and off-screen lines are separated.

Lip-motion analysis · Voice pitch & identity analysis · Active-speaker detection

3

Dubbing with voice cloning

The whole episode is dubbed in one continuous pass, so each character's cloned voice stays consistent from scene to scene. Levels are matched to the original dialogue.

Voice cloning · Neural dubbing · Loudness matching

4

Script verification & correction

Both the original and the English are transcribed and compared line by line. Wrong words, names or meanings are rebuilt in the character's own cloned voice, down to single-word edits.

Speech recognition · Forced word alignment · Dedicated voice models

5

Multilingual detection

Language identification runs on every spoken phrase. It found the Serbian-language scenes, which were then transcribed in Serbian and translated directly.

Language identification · Serbian → English translation

6

Lip sync

Each on-camera line is lip-synced on a shot-by-shot basis and pinned to the speaker's face. The model only hears that speaker's own words, so listeners' mouths stay still.

Lip-sync model · Face pinning · Per-speaker audio gating · Dual-pass two-shots

7

Frame-exact assembly

Lip-synced shots are cut back into the original picture on exact frame boundaries. Everything else stays original.

Frame-accurate conform · Mastering-grade encode

8

Verification & delivery

Every file is checked for frame count, format, audio-to-picture alignment and spoken language before delivery.

Automated QA · Visual review

03 — STAGE 1 · SCENE, SHOT & FACE ANALYSIS

We read every frame before changing any of them

All 8,874 frames of the episode were analyzed. The workflow detects every cut, finds every face, tracks it through the shot and decides which face is speaking. Only that face is ever lip-synced.



Face detection across the episode. Every face is located with a confidence score and five anchor points (eyes, nose, mouth corners). The face marked **active speaker** is the one whose mouth matches the dialogue; listeners and background faces are left untouched.



Dense landmark mesh. For each tracked face, a 478-point mesh measures lip opening and jaw movement in every frame, even in profile and low light.

8,874
frames scanned for faces, cuts and mouth movement

52
speaker anchor points placed on exact frames and verified against face detection

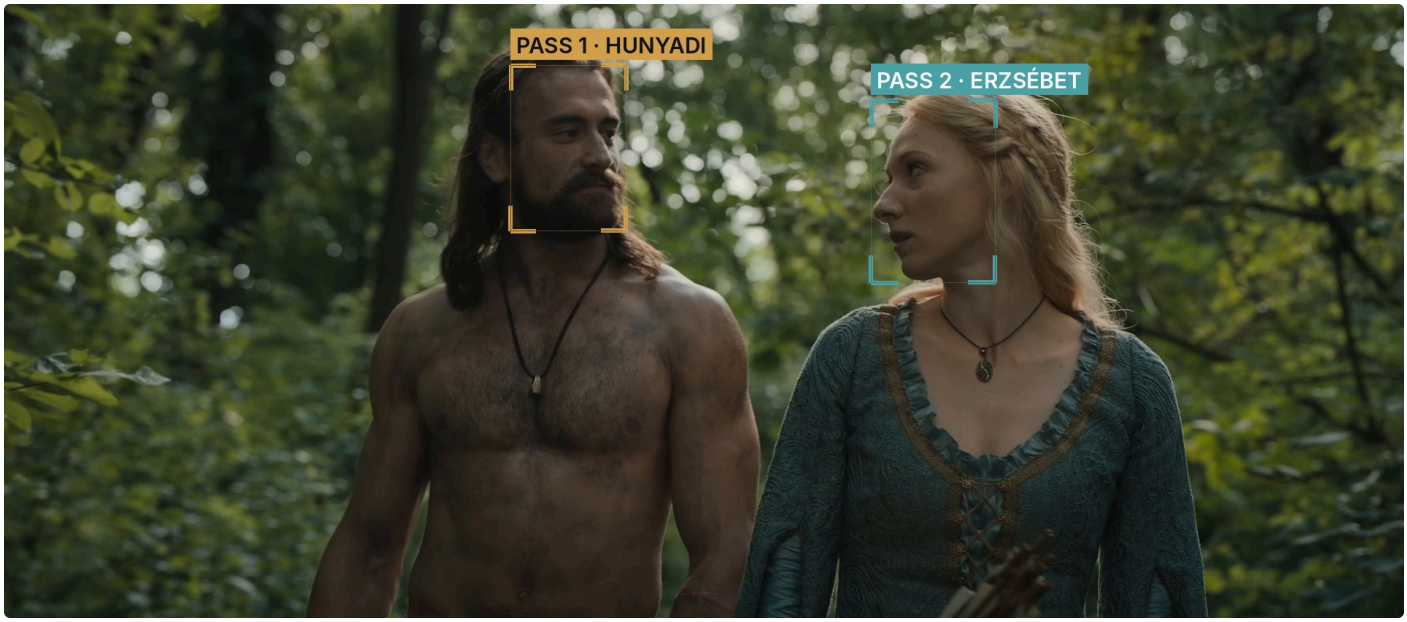
1 frame
precision on every cut and every lip-sync boundary (1/24 s)

Shots where the speaker is turned away, hidden behind a hand, too small in a wide shot or simply off-screen are flagged as **audio only**. They get the English voice while their picture stays exactly as shot.

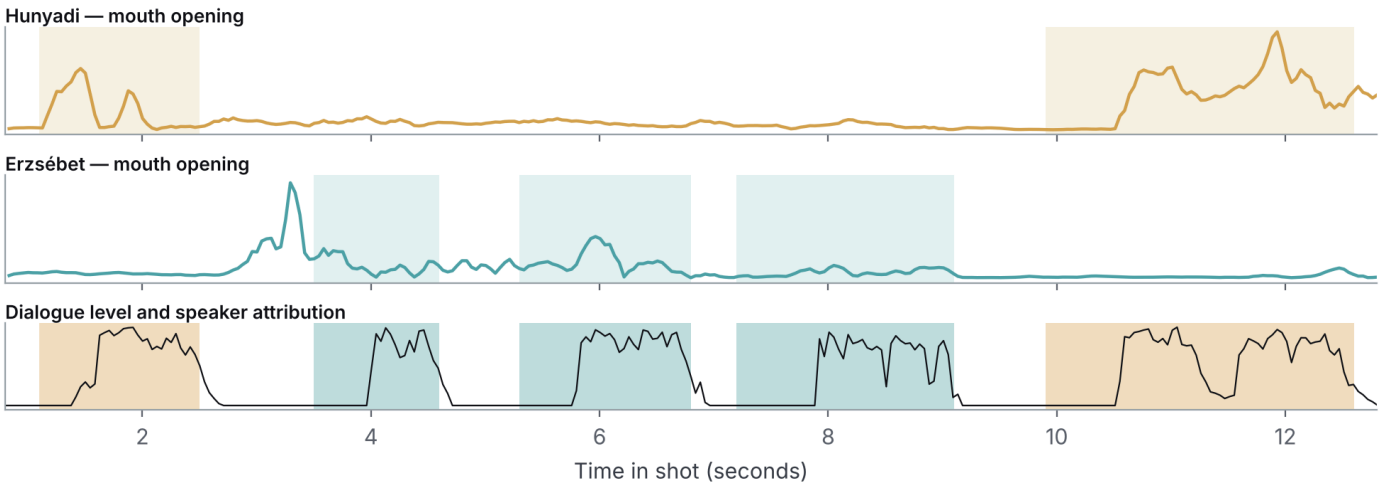
04 — STAGE 2 · SPEAKER IDENTIFICATION

Who is speaking, frame by frame

A lip-sync model will animate whatever face it is given. The precision comes from deciding, line by line, whose mouth should move and whose should not.



A two-person shot (Hunyadi and Erzsébet in the forest, 17 seconds). Each character is lip-synced in a separate pass, and each pass receives only that character's own lines.



Measured mouth opening for both faces against the dialogue. Gold and teal bands show which lines the workflow attributed to Hunyadi and to Erzsébet. Attribution combines mouth movement with voice analysis of every sentence (pitch and voice identity).

Lip-motion analysis

Mouth opening is tracked per face and correlated with the speech in the dialogue track.

Voice analysis

Pitch and voice-identity analysis of each sentence separate male, female and child speakers, and tell characters apart.

Word-level gating

Each lip-sync pass hears only its speaker's words, timed to the individual word, so a reply never moves the wrong mouth.

05 — CHARACTERS & VOICE CLONES

20+ voices, each cloned from its own actor

20+
speaking roles identified in the episode

100%
of roles voiced from the original actor's own performance

6
principal characters with dedicated voice models

Every speaking role is voiced with a clone of the actor who played it, so the English keeps each performer's timbre, age and energy. The six principal characters also received a **dedicated voice model**, built from clean, isolated takes of their performance. We use these models to re-voice individual lines and words with full control.

Principal cast · dedicated voice models



Hunyadi János
Lead · the hero

DEDICATED VOICE MODEL

Model sample **9.7 s** Clips **7**



Szilágyi Erzsébet
Lead · Hunyadi's love interest

DEDICATED VOICE MODEL

Model sample **11.9 s** Clips **6**



Branković
Serbian lord · speaks Serbian

DEDICATED VOICE MODEL

Model sample **10.5 s** Clips **5**



Branković's daughter
Serbian court · speaks Serbian

DEDICATED VOICE MODEL

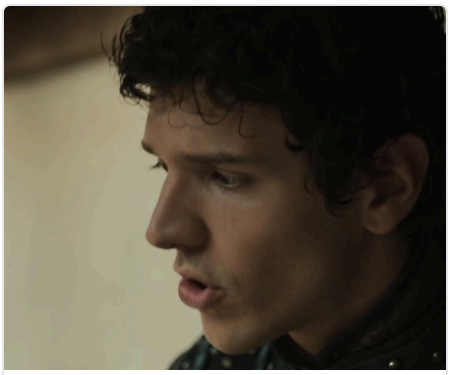
Model sample **7.3 s** Clips **5**



Jankó's father
Prologue · the attack

DEDICATED VOICE MODEL

Model sample **13.2 s** Clips **5**



Miklós
Erzsébet's betrothed

DEDICATED VOICE MODEL

Model sample **6.5 s** Clips **3**

05 — CHARACTERS & VOICE CLONES (CONTINUED)

Supporting cast

Supporting and minor roles are voiced with performance clones made from their own lines in the episode. The same voice follows the character from scene to scene.



The archery master
Prologue · 4 clips
PERFORMANCE CLONE



Young Jankó
Hunyadi as a boy · 2 clips
PERFORMANCE CLONE



Jankó's mother
Prologue · 2 clips
PERFORMANCE CLONE



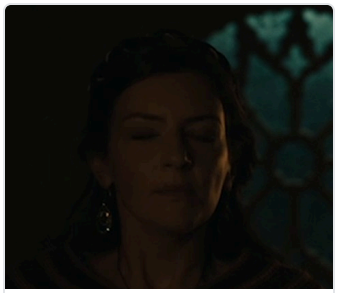
The Queen
Royal banquet
PERFORMANCE CLONE



Nobleman in furs
Royal banquet · 2 clips
PERFORMANCE CLONE



The courtier
Court scenes · 4 clips
PERFORMANCE CLONE



Lady of the hall
Castle corridor
PERFORMANCE CLONE



Lord with the gold chain
Balcony scene
PERFORMANCE CLONE

All speaking roles

CHARACTER	LANGUAGE	VOICE	SCENES
Hunyadi János	Hungarian, Serbian	DEDICATED MODEL	Forest, Branković's court, banquet
Szilágyi Erzsébet	Hungarian	DEDICATED MODEL	Forest, castle, banquet
Branković · his daughter	Serbian	DEDICATED MODELS	Branković's court, banquet
Jankó's father · Miklós	Hungarian	DEDICATED MODELS	Prologue · castle balcony
Archery master, young Jankó, mother	Hungarian	PERFORMANCE CLONES	Prologue: archery lesson, the attack
Queen, nobleman in furs, courtier, lords and ladies	Hungarian	PERFORMANCE CLONES	Court, corridor, royal banquet
Guards, servants, villagers, off-screen voices	Hungarian, Serbian	PERFORMANCE CLONES	Throughout

06 — STAGES 3 & 4 · DUBBING, SCRIPT VERIFICATION & CORRECTION

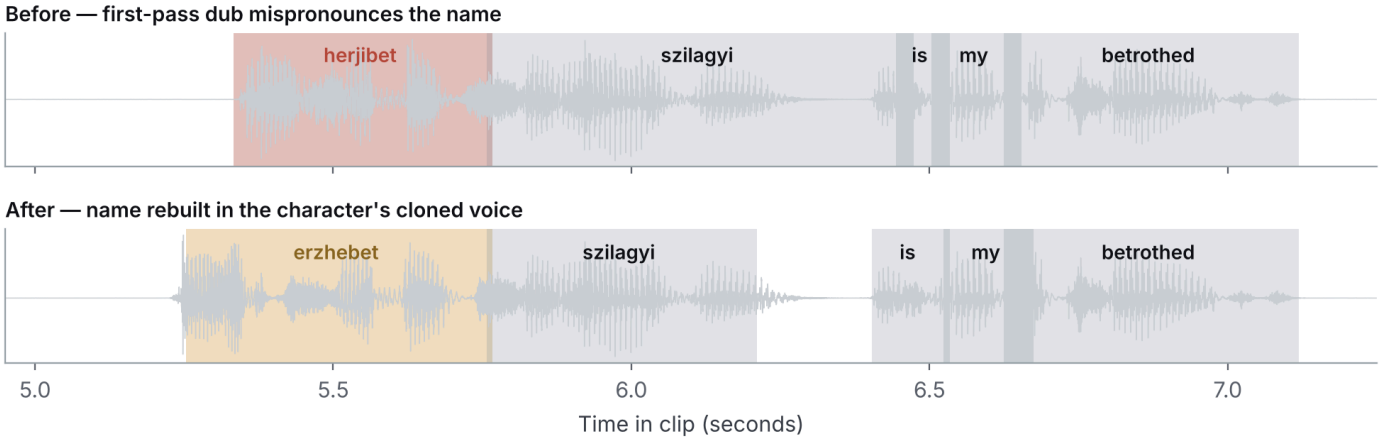
Line-by-line verification, down to a single word

Dubbing with voice cloning

The episode is dubbed as one continuous performance rather than clip by clip. Each character keeps the same cloned voice throughout, and the English is level-matched to the original dialogue so the mix sits exactly where the original did.

Script verification

We transcribe both the original dialogue and the English dub, then compare them line by line. Mistranslated lines, mispronounced names and missing lines are corrected in the character's own dedicated voice model, timed to the actor's original delivery.



Word-level surgery. Forced alignment locates every word to the millisecond. Here only the mispronounced name is rebuilt in the character's cloned voice, and the rest of the actor-matched line is kept untouched.

Examples of corrections

CLIP	FIRST-PASS DUB	DELIVERED ENGLISH	CORRECTION
90527	*"Bring me my partridge..."*	"Bring me my **bracer**, then do what your mother says, understand?"	Period vocabulary · single word
102229	*"You're Miho's sister..."*	"You're **Mihály's** sister, little Erzsébet Szilágyi."	Character name · single word
104461	*"King Kasnik usually plays with his cow."*	"The king is late, as usual. No doubt still amusing himself with his whores."	Meaning · full line re-voiced
105002	*"will be want you"*	"You'll behave, won't you?"	Garbled line · re-voiced
104695	*(line missing)*	"It'll be fine."	Missing line · inserted
108806	*"...as jealous as we lucky."*	"Not nearly as jealous as **Újlaki**."	Character name · single word

36
targeted corrections across the episode

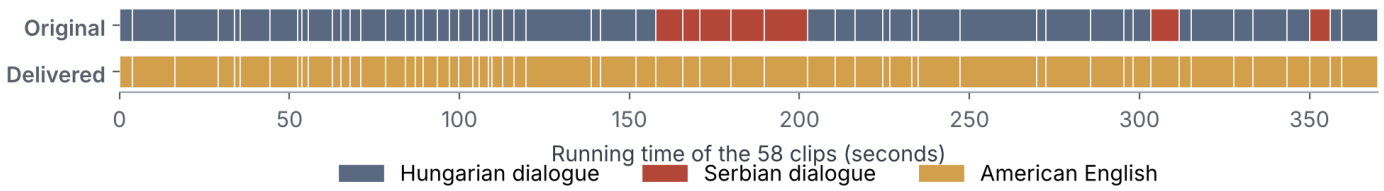
0.93–1.12×
pace range of re-voiced lines: natural speed, never rushed

100%
of corrected lines re-checked by transcription after editing

07 — STAGE 5 · MULTILINGUAL DETECTION

Two source languages, one English episode

Language identification ran on every one of the 137 spoken phrases in the finished episode. It confirmed that no Hungarian dialogue remains (only character names). It also showed that the Serbian court is performed in Serbian, so those scenes were transcribed in Serbian and translated directly from Serbian into English.



Language map of the episode. Top: the original dialogue, Hungarian (slate) and Serbian (red). Bottom: the delivered episode, entirely American English.

Serbian scenes · translated from Serbian

CLIP	CHARACTER	ORIGINAL (SERBIAN)	DELIVERED ENGLISH
103835	**Branković**	*Taj vuk pripada meni, jer si ga ubio na srpskoj zemlji.*	That wolf belongs to me. You killed it on Serbian soil.
104123	**Hunyadi**	*Ali sam ga dao tvojoj kćeri.*	But I gave it to your daughter.
104240	**Daughter**	*Pa kako to izgledaš? Ajde, umij se, obuci se.*	Look at the state of you! Wash up, get dressed!
104240	**Branković**	*Kralj dolazi. Tvoj, mađarski.*	The king is coming. Your Hungarian king.
104461	**Branković**	*Oprosti, kćeri. Janko!*	Excuse me, daughter. Janko!
104695	**Branković**	*Slušaj, ovo veče je za mene jako važno.*	Listen, this evening is very important to me.
104695	**Branković**	*Ako razljutiš kralja, u ovu kuću više ne dolaziš.*	If you anger the king, don't you ever come back to this house.
107827	**Branković**	*Mara mi je potrebna da živi uz mene, pomaže mi u svemu, ne mogu je dati na tvoj dvor.*	I need Mara here with me. She helps me with everything. I can't give her to your court.
109692	**Servant**	*Gospodaru!*	My lord!

08 — REFERENCE SCRIPT

The English script, scene by scene

Excerpts of the bilingual reference script, with the original line beside the English as delivered. Clip numbers refer to the delivered files.

Prologue · The archery lesson

Clips 87434–88693 · Hungarian

CLIP	CHARACTER	ORIGINAL	ENGLISH
87434	**Archery master**	*Jankó, gyere ide!*	Jankó, come here.
87646	**Archery master**	*Na, mutasd! Hogy tartod azt az íjat?*	Now let me see how you hold that bow.
87646	**Archery master**	*Az ideget tartsd távolabb, megüti az arcod. Engedd lejjebb a vállad.*	Keep the string further away or it'll slap the cheek. Drop your shoulder.
88011	**Archery master**	*Ne a kezeddel célozz. Ne is a szemeddel. Itt.*	Don't aim with your hand, nor with your eye. Right here.
88514	**Mother**	*Jankó, hányszor mondtam, hogy ne az udvaron játssz azzal az íjjal?*	Jankó, how many times have I told you not to play with that bow in the courtyard?

Prologue · The attack

Clips 90058–90694 · Hungarian

CLIP	CHARACTER	ORIGINAL	ENGLISH
90058	**Father**	*Ide figyeljetek! El kell bújnotok!*	Listen to me. You need to hide.
90058	**Father**	*Bármi történik, azt teszitek, amit anyátok mond, megértettétek?*	Whatever happens, do as your mother says. Do you hear me?
90252	**Young Jankó**	*Én maradok veled.*	I'm staying with you.
90527	**Father**	*Ide figyelj, a te dolgod most az, hogy ezt túléld.*	Listen to me. Your only job right now is to survive this.
90694	**Father**	*Nem lesz semmi baj. Minden rendben lesz.*	It'll be alright. Everything will be fine.

The forest · Hunyadi meets Erzsébet

Clips 102065–103629 · Hungarian

CLIP	CHARACTER	ORIGINAL	ENGLISH
102065	**Erzsébet**	*Most már a tiéd lehet.*	Well, now it's all yours.
102148	**Hunyadi**	*Segítesz visszajutni a várba?*	Will you help me find my way back to the castle?
102229	**Hunyadi**	*Hunyadi János vagyok.*	I'm János Hunyadi.
102229	**Erzsébet**	*Tudom, ki vagy. Minden lány tudja a neved. A bátyámmal szoktál mulatni.*	I know who you are. Every girl knows your name. You carouse with my brother.
102229	**Hunyadi**	*Te Mihály húga vagy? A kis Szilágyi Erzsébet? Mit keresel itt?*	You're Mihály's sister, little Erzsébet Szilágyi. What are you doing out here?
102229	**Erzsébet**	*A királyné frajláival jöttem.*	I came with the queen's ladies.
103629	**Hunyadi**	*De valójában ez téged illet, szerelmem.*	But in truth, it belongs to you, my love.

08 — REFERENCE SCRIPT (CONTINUED)

Branković's court

Clips 103835–104695 · Serbian

CLIP	CHARACTER	ORIGINAL	ENGLISH
103835	**Branković**	*Hej, Janko! Taj vuk pripada meni, jer si ga ubio na srpskoj zemlji. Dođi.*	Hey, Janko! That wolf belongs to me. You killed it on Serbian soil. Come on!
104123	**Hunyadi**	*Ali sam ga dao tvojoj kćeri.*	But I gave it to your daughter.
104240	**Daughter**	*Pa kako to izgledaš? Ajde, umij se, obuci se.*	Look at the state of you! Wash up, get dressed!
104240	**Branković**	*Kralj dolazi. Tvoj, mađarski.*	The king is coming. Your Hungarian king.
104240	**Hunyadi**	*Je l' Zigismund već tu?*	Is Sigismund here already?
104461	**Branković**	*Samo kraljica. Kralj kasni kao obično, sigurno se još zabavlja sa svojim kurvama.*	Only the queen. The king is late, as usual. No doubt still amusing himself with his whores.
104461	**Daughter**	*Oče!*	Father!
104461	**Branković**	*Oprosti, kćeri. Janko!*	Excuse me, daughter. Janko!
104695	**Branković**	*Slušaj, ovo veče je za mene jako važno. Ako razljutiš kralja, u ovu kuću više ne dolaziš. Zato pazi što radiš.*	Listen, this evening is very important to me. If you anger the king, don't you ever come back to this house. So watch what you do.
104695	**Daughter**	*Biće dobro.*	It'll be fine.

The royal banquet

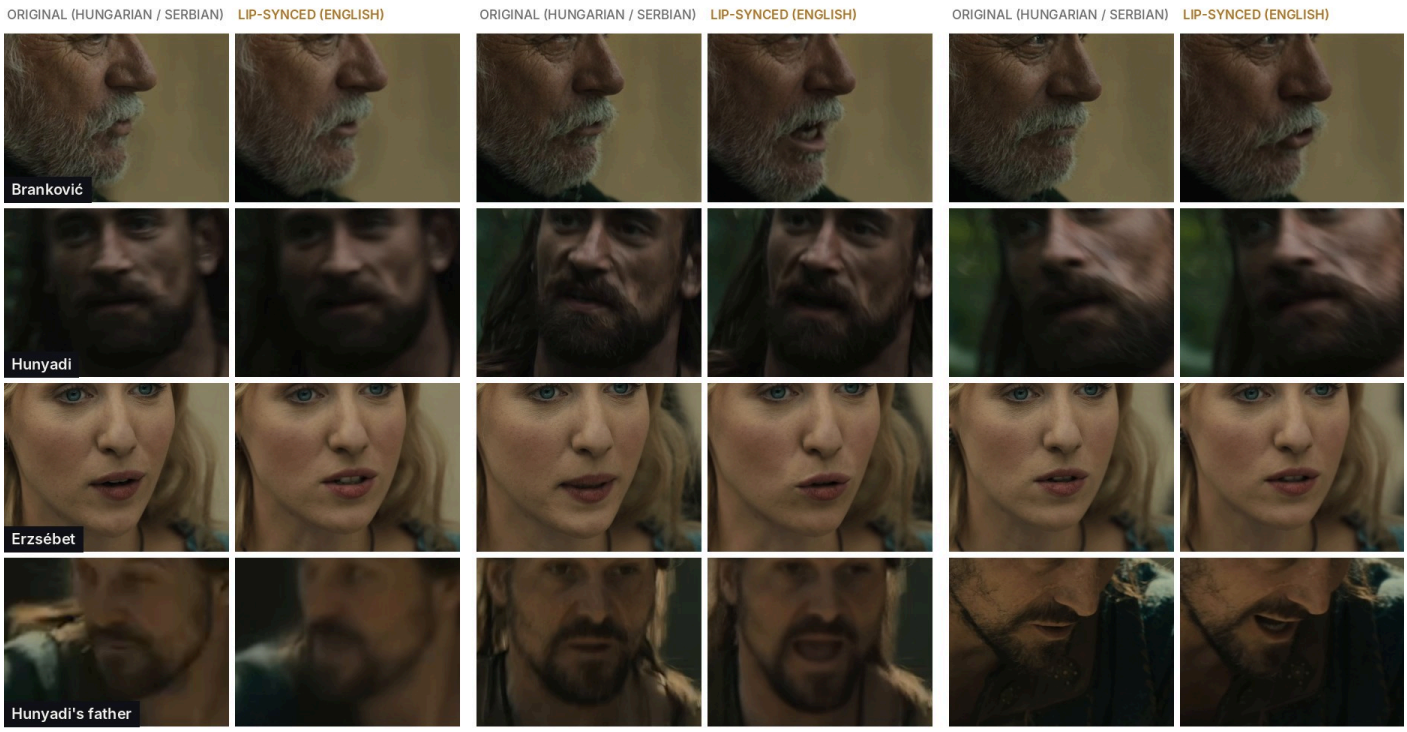
Clips 107025–108806 · Hungarian & Serbian

CLIP	CHARACTER	ORIGINAL	ENGLISH
107025	**Nobleman**	*Éppen drága Brankovicsot faggatom, hogy mi a terve a kisebbik lányával.*	I was just pressing dear Branković about what he plans for his younger daughter.
107339	**Nobleman**	*Gondoskodtál már a lányod jövőjéről, Brankovics? Beállhatna a királyné mellé udvarhölgynek.*	Have you seen to your daughter's future yet, Branković? She could join the Queen as a lady-in-waiting.
107575	**Courtier**	*Ez egészen pompás ötlet, felség.*	A splendid idea, Your Majesty.
107637	**The Queen**	*A frajláimat én választom, a kurváidat te.*	I choose my ladies-in-waiting, and you choose your whores.
107827	**Branković**	*Mara mi je potrebna da živi uz mene, pomaže mi u svemu, ne mogu je dati na tvoj dvor.*	I need Mara here with me. She helps me with everything. I can't give her to your court.
108382	**Erzsébet**	*Már megint te? Pont olyan beképzelt vagy, mint ahogy gondoltam. De legalább jól táncolsz.*	You again? You're just as arrogant as I thought. At least you can dance.
108806	**Hunyadi**	*Annyira biztos nem, mint Újlaki.*	Not nearly as jealous as Újlaki.

09 — STAGE 6 · LIP SYNC

The English on the actor's lips

Our lip-sync model re-renders only the lower face of the speaking actor, shot by shot, so the mouth forms the English words. Skin, beard, lighting, grain and the rest of the frame stay as photographed.



Original vs. lip-synced, same frames. Three moments per character, taken from the delivered episode. The original mouth shapes follow the Hungarian or Serbian; the lip-synced ones follow the English.

Shot-trimmed

Each pass covers exactly one shot, cut on the frame, so a lip-sync never runs across an edit.

Face-pinned

Each pass is anchored to the speaker's face on a chosen frame, verified against face detection.

Speaker-gated

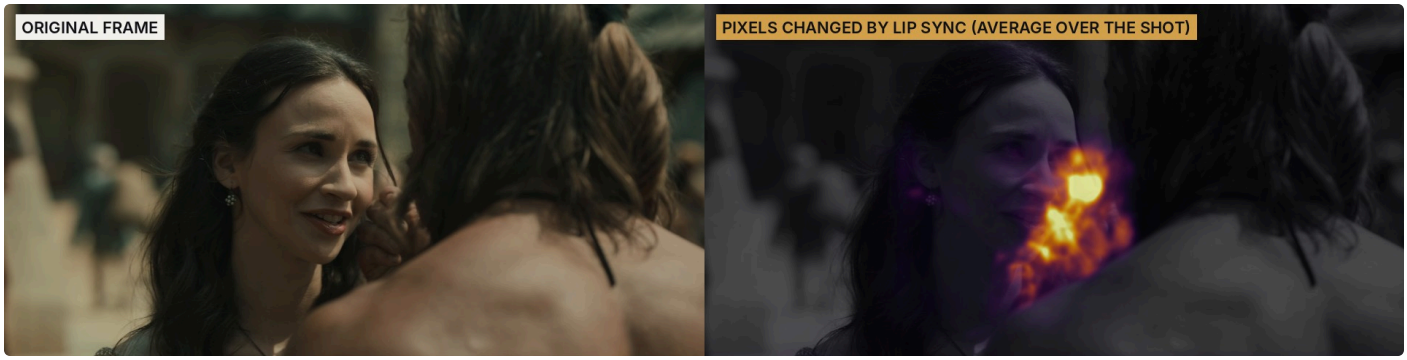
The model hears only the on-screen speaker's own words, so listeners don't move their lips.

Dual-pass two-shots

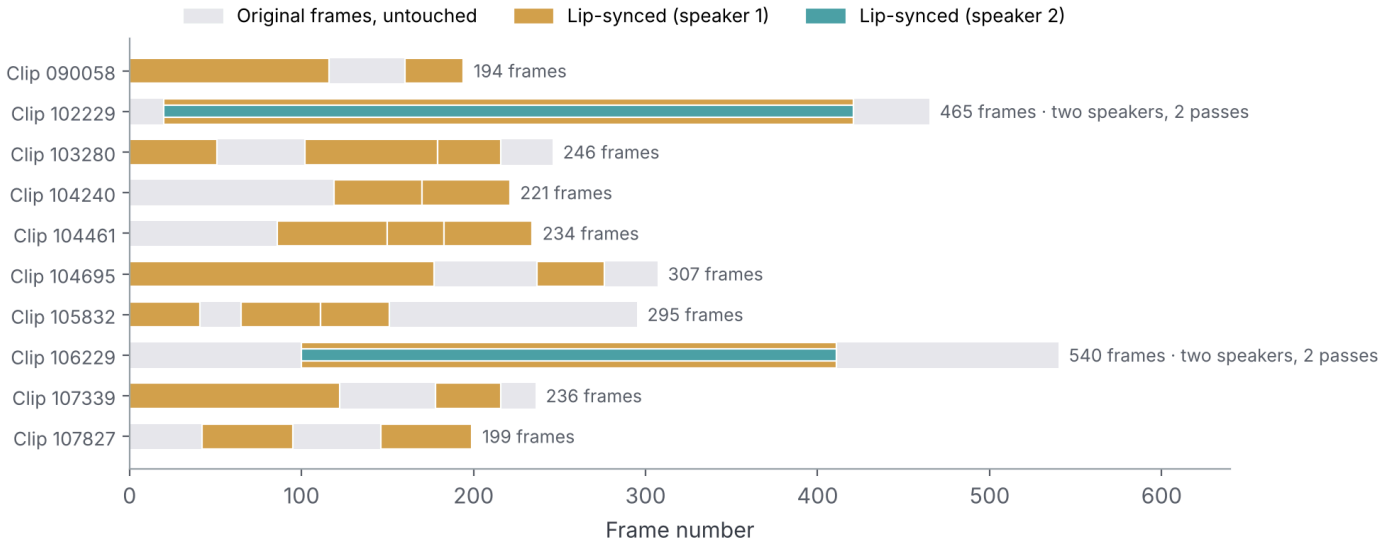
When two people talk in one shot, each gets their own pass, chained on top of the other.

10 — STAGES 6 & 7 · PRECISION & FRAME-EXACT ASSEMBLY

Change only what must change



Where the picture was altered. Average difference between the original and the lip-synced shot. The change is confined to the speaking actor's mouth and jaw; the listener, background and lighting are untouched.



Frame-exact assembly. Each clip is rebuilt on exact frame boundaries: lip-synced shots (gold, and teal for the second speaker in a two-shot) are cut into the untouched original frames (grey). Ten of the 38 lip-synced clips shown.

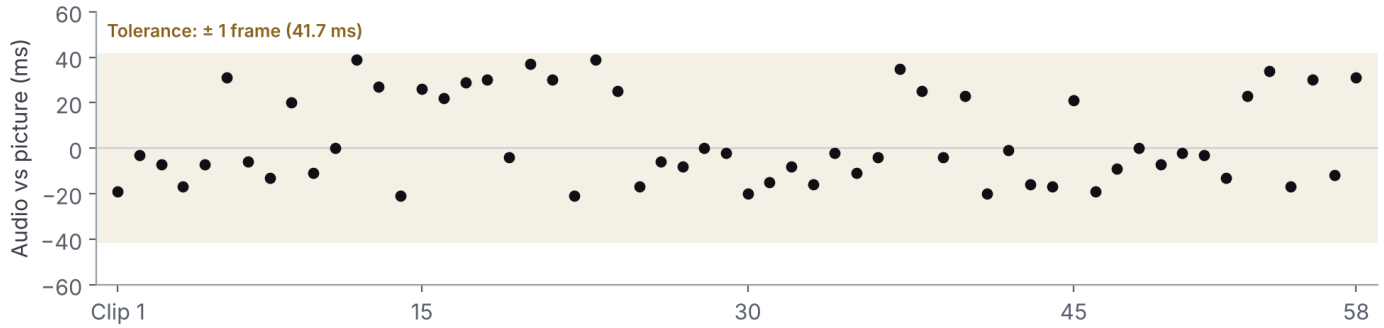
38
clips with lip sync · 20 clips with picture untouched

52
shot-level lip-sync passes, each frame-checked

0
frames added, dropped or shifted against the source

11 — STAGE 8 · VERIFICATION & DELIVERY

Every file verified before it leaves



Audio-to-picture alignment of all 58 delivered clips. Every clip falls well inside a one-frame tolerance.

Quality checks

- ✓ **Frame count** identical to the source, all 58 clips
- ✓ **Format** 1920×1080, 23.976 fps, as delivered to us
- ✓ **Audio** one English dialogue track, AAC 48 kHz
- ✓ **Sync** audio within one frame of the picture
- ✓ **Content** each file carries the verified English track
- ✓ **Language** all 137 spoken phrases identified as English (names excepted)
- ✓ **Lip sync** every pass checked: the change sits on the pinned face, mouth crops reviewed by eye

Deliverables

58 video files, one per original clip, in the original order. Each keeps its original file name with an **_EN** suffix:

```
RISE_OF_THE_RAVEN_S01E01_…_Splits00087434_EN.mp4
```

Picture is the original master wherever no lip sync was needed, and frame-exact lip-synced picture elsewhere, with the English dialogue track.

READY FOR REVIEW

A native-speaker pass on the final English is always welcome; our reference script makes that review quick.

DUBBING · VOICE CLONING · LIP SYNC · LOCALIZATION